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Lab 01

SloshMPend

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0. Introduction

Lateral sloshing refers to the oscillatory motion of a liquid in a partially filled container subjected to horizontal accelerations.

This phenomenon is characterized by the formation of waves that alternately rise and fall along the container sidewalls which causes the tank center of mass oscillation.

Fluid sloshing inside a partially filled tank is a critical phenomenon for all types of structures: the forces and moments generated by the sloshing mass can significantly affect the dynamics and stability of any aerospace vehicle.

In this report, we aim to analyze the problem and simulate the sloshing of a fluid mass of approximately 785.40 kg inside a cylindrical tank with a diameter of 1.0 m and unspecified height.

This analysis refers the approach introduced in the study “Sloshing Behavior of Liquids in Moving Containers” by Franklin T. Dodge of the Southwest Research Institute, San Antonio, Texas. [1]

This method simplifies the complex sloshing dynamics by considering only the lowest-order sloshing modes, which are the most influential and representative in practice. This simplification allows for better control and prediction of the interaction between fluid and structure and its effects on vehicle dynamic. The core idea of this approach is to model the phenomenon through an equivalent mechanical system composed of a set of pendulums. These pendulums oscillate due to the tank acceleration, and their natural frequencies expressed as $\sqrt{g/L}$ depend on the changing acceleration, like the natural sloshing frequency of the liquid inside the tank. This analogy enables a dynamic and physically consistent representation of the sloshing phenomenon under different motion conditions.

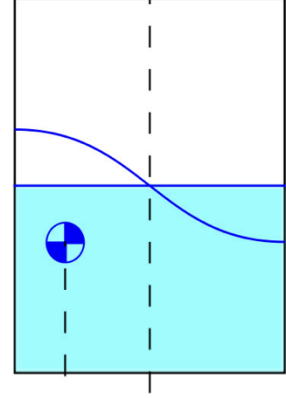


Figure 1: Schematic illustration of a slosh wave

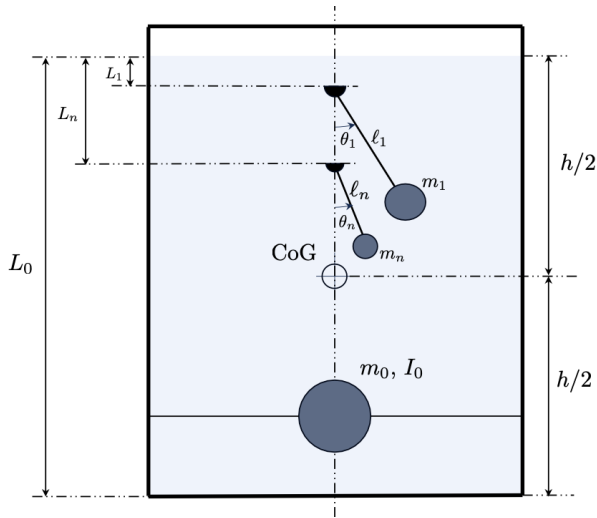
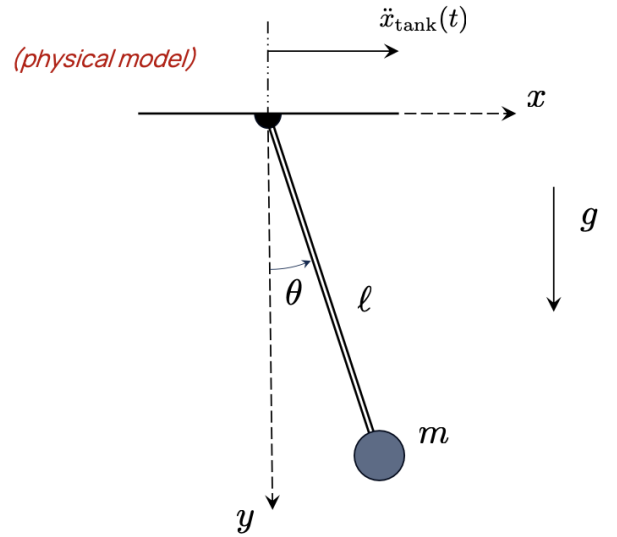


Figure 2: Mechanical model of sloshing



1. Task 1: Parameters of the multi-pendulum model

According to an equivalent mechanical model of the sloshing dynamics of the cylindrical tank, the formulas we used to calculate the parameters are as follows, shown in Table 3.2 from Dodge's work [1],

$$m_{\text{liq}} = (\pi/4)\rho d^2 h \quad \xi_1 = 1.841; \quad \xi_2 = 5.329; \quad \xi_3 = 8.531; \quad \xi_n \approx \xi_{n-1} + \pi$$

Description	Formula
Slosh mass, m_n	$m_{\text{liq}} \left[\frac{d \tanh(2\xi_n h/d)}{\xi_n (\xi_n^2 - 1)h} \right]$
Pendulum length, L_n	$\frac{d}{2\xi_n \tanh(2\xi_n h/d)}$
Pendulum hinge location, H_n	$\frac{h}{2} - \frac{d}{2\xi_n} \left[\tanh(\xi_n h/d) - \frac{1}{\sinh(2\xi_n h/d)} \right]$
Rigidly attached mass, m_0	$m_{\text{liq}} - \sum m_n \approx m_{\text{liq}} \left[1 - \frac{d \tanh(2\xi_n h/d)}{\xi_n (\xi_n^2 - 1)h} \right]$
Rigidly attached mass location, H_0	$\sum m_n (H_n - L_n) / m_0$

Table 1: Table extracted from [1], p.48.

Using these formulas, we can find the values of mass, length, pulsation, frequency, and height of n pendulums (n = 10):

	m (kg)	L (m)	w (rad/s)	f (Hz)	H (m)
m_1	178.3000	0.2719	6.0062	0.1665	0.2554
m_2	5.3792	0.0938	10.2252	0.0978	0.4062
m_3	1.2826	0.0586	12.9375	0.0773	0.4414
m_4	0.4975	0.0428	15.1333	0.0661	0.4572
m_5	0.2427	0.0338	17.0486	0.0587	0.4662
m_6	0.1361	0.0278	18.7695	0.0533	0.4722
m_7	0.0838	0.0237	20.3453	0.0492	0.4763
m_8	0.0552	0.0206	21.8075	0.0459	0.4794
m_9	0.0383	0.0183	23.1777	0.0431	0.4817
m_{10}	0.0277	0.0164	24.4713	0.0409	0.4836

Table 2: Value of the parameters

2. Task 2: EOM and state-space model derivation

2.1. Task 2a: EOM and state-space model

To study the movement of the system we begin by writing the Lagrangian equation based on the D.o.F. θ for the system:

$$\frac{d}{dt} \left(\frac{\partial \mathcal{L}_i}{\partial \dot{\theta}_i} \right) - \frac{\partial \mathcal{L}_i}{\partial \theta_i} = Q_i \quad (1)$$

Where:

$$\mathcal{L}_i = \frac{1}{2} m_i \left(l_i^2 \cos^2 \theta \dot{\theta}^2 + l_i^2 \sin^2 \theta \dot{\theta}^2 \right) - m_i g l_i \cos \theta_i$$

$$= \frac{1}{2} m_i \left(l_i^2 \dot{\theta}^2 \right) - m_i g l_i \cos \theta_i$$

Calculating the different components of the equation:

$$\frac{\partial \mathcal{L}}{\partial \dot{\theta}} = m_i l_i^2 \dot{\theta}$$

$$\frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{\theta}} \right) = m_i l_i^2 \ddot{\theta}$$

$$\frac{\partial \mathcal{L}}{\partial \theta} = -m_i g l_i \sin \theta$$

We can express the generalized force component Q_i through the Principle of Virtual Displacements:

$$\delta \mathcal{L} = \mathbf{F} \cdot \delta \mathbf{p} = m_i \ddot{x}_{\text{tank}} (-\delta x_{m_i}) = -m_i \ddot{x}_{\text{tank}} l \cos \theta_i \delta \theta_i$$

where $\delta x_{m_i} = l \cos \theta_i \delta \theta_i$

$$\delta \mathcal{L} = Q_i \delta \theta_i \quad \Rightarrow \quad Q_i = -m_i l \cos \theta_i \ddot{x}_{\text{tank}}$$

The total equation becomes:

$$m_i l_i^2 \ddot{\theta}_i + m_i g l_i \sin \theta_i = -m_i l_i \cos \theta_i \ddot{x}_{\text{tank}}$$

simplifying the equation we get:

$$\ddot{\theta}_i = -\frac{g}{l_i} \sin \theta_i - \frac{1}{l_i} \cos \theta_i \ddot{x}_{\text{tank}} \quad (2)$$

Adding the initial conditions:

$$\begin{cases} \theta_i(t_0) = \theta_0 \\ \dot{\theta}_i(t_0) = \omega_0 \end{cases} \quad i = 1, \dots, 10$$

We now want to switch to the state-space formulation:

$$\mathbf{X} = \begin{Bmatrix} \theta_1 \\ \dot{\theta}_1 \\ \theta_2 \\ \dot{\theta}_2 \\ \vdots \\ \theta_{10} \\ \dot{\theta}_{10} \end{Bmatrix} = \begin{Bmatrix} X_{1,1} \\ X_{1,2} \\ X_{2,1} \\ X_{2,2} \\ \vdots \\ X_{10,1} \\ X_{10,2} \end{Bmatrix}$$

input: $u = \ddot{x}_{\text{tank}}$

$$\dot{\mathbf{X}} = \begin{Bmatrix} \dot{\theta}_1 \\ \ddot{\theta}_1 \\ \dot{\theta}_2 \\ \ddot{\theta}_2 \\ \vdots \\ \dot{\theta}_{10} \\ \ddot{\theta}_{10} \end{Bmatrix} = \begin{Bmatrix} \dot{X}_{1,1} \\ \dot{X}_{1,2} \\ \dot{X}_{2,1} \\ \dot{X}_{2,2} \\ \vdots \\ \dot{X}_{10,1} \\ \dot{X}_{10,2} \end{Bmatrix} = \begin{Bmatrix} X_{12} \\ -\frac{g}{\ell_1} \sin X_{1,1} - \frac{1}{\ell_1} \cos X_{1,1} u \\ X_{22} \\ -\frac{g}{\ell_2} \sin X_{2,1} - \frac{1}{\ell_2} \cos X_{2,1} u \\ \vdots \\ X_{102} \\ -\frac{g}{\ell_{10}} \sin X_{10,1} - \frac{1}{\ell_{10}} \cos X_{10,1} u \end{Bmatrix}$$

We want to downscale the system from a second order one to a first order one:

$$\begin{cases} \dot{x}_{i_1} = f_{i_1}(x_i, u, \mathbf{p}) = x_{i_2} \\ \dot{x}_{i_2} = f_{i_2}(x_i, u, \mathbf{p}) = -\frac{g}{l_i} \sin x_{i_1} - \frac{1}{l_i} \cos x_{i_1} u \\ x_i(t_0) = x_{t_0} \end{cases}$$

Output: *the net force F_x exerted on the tank along x -direction*

$$y = g(\dot{x}_i, u, \mathbf{p}) = -m_0 u - \sum_i^m m_i (u + \dot{x}_{i,2} l_i \sin x_{i,1}) \quad (3)$$

Omitting time dependency and vector of parameters

$$\begin{cases} \dot{x} = f(x, u) \\ y = g(x, u) \end{cases}$$

Equilibrium points:

$$x_{eq} \Rightarrow f(x_{eq}, 0) = 0 \Rightarrow \dot{x} = 0 \Rightarrow u_{eq} = 0$$

$$\Rightarrow \begin{cases} x_{i,2,eq} = 0 \\ -\frac{g}{l_i} \sin x_{i,1,eq} = 0 \end{cases} \Rightarrow \begin{cases} x_{i2,eq} = 0 \\ \sin x_{i,1,eq} = 0 \end{cases}$$

$$x_{eq} = \begin{Bmatrix} n_i \pi \\ 0 \end{Bmatrix}, \quad m_i = 1, 2, \dots \quad i = 1, \dots, 10$$

We want to consider: $n_i = 0$

$$x_{eq} = \begin{Bmatrix} 0 \\ 0 \end{Bmatrix}$$

Linearization

$$\Delta x = x - x_{eq}, \quad \Delta u = u - 0 = u, \quad \Delta y = y - g(x_{eq}, 0)$$

$$\begin{cases} \Delta \dot{x}(t) = \left. \frac{\partial f}{\partial x} \right|_{x_{eq}} \Delta x + \left. \frac{\partial f}{\partial u} \right|_{x_{eq}} \Delta u \\ \Delta y(t) = \left. \frac{\partial g}{\partial x} \right|_{x_{eq}} \Delta x + \left. \frac{\partial g}{\partial u} \right|_{x_{eq}} \Delta u \end{cases}$$

$$\Rightarrow \begin{cases} \Delta \dot{x}(t) = A \Delta x + B \Delta u \\ \Delta y(t) = C \Delta x + D \Delta u \end{cases}$$

So the linearized matrices will have sizes:

$$A = (20 \times 20), \quad B = (20 \times 1), \quad C = (1 \times 20), \quad D = (1 \times 1)$$

Where

$$\begin{aligned} A &= \left[\begin{array}{ccc} \frac{\partial f_{1,1}}{\partial x_{1,1}} & \dots & \frac{\partial f_{1,1}}{\partial x_{10,2}} \\ \frac{\partial f_{1,2}}{\partial x_{1,1}} & \dots & \frac{\partial f_{1,2}}{\partial x_{10,2}} \\ \vdots & \ddots & \vdots \\ \frac{\partial f_{10,2}}{\partial x_1} & \dots & \frac{\partial f_{10,2}}{\partial x_{10}} \end{array} \right] \Big|_{x_{eq}} \\ B &= \left[\begin{array}{c} \frac{\partial f_{1,1}}{\partial u} \\ \frac{\partial f_{1,2}}{\partial u} \\ \vdots \\ \frac{\partial f_{10,2}}{\partial u} \end{array} \right] \Big|_{x_{eq}} \\ C &= \left[\frac{\partial g}{\partial x_{1,1}} \quad \frac{\partial g}{\partial x_{1,2}} \quad \dots \quad \frac{\partial g}{\partial x_{10,1}} \quad \frac{\partial g}{\partial x_{10,2}} \right] \Big|_{x_{eq}} \\ D &= \left[\frac{\partial g}{\partial u} \right] \Big|_{x_{eq}} \end{aligned}$$

After some calculations and the linearization we get the definitive state-space matrices for the undamped model:

$$\begin{aligned} A &= \begin{bmatrix} 0 & 1 & 0 & 0 & \dots & 0 & 0 \\ -\omega_1^2 & 0 & 0 & 0 & \dots & 0 & 0 \\ 0 & 0 & 0 & 1 & \dots & 0 & 0 \\ 0 & 0 & -\omega_2^2 & 0 & \dots & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & 0 & \dots & 0 & 1 \\ 0 & 0 & 0 & 0 & \dots & -\omega_{10}^2 & 0 \end{bmatrix} \\ B &= \begin{bmatrix} 0 \\ -\frac{1}{l_1} \\ \vdots \\ 0 \\ -\frac{1}{l_{10}} \end{bmatrix} \\ C &= \begin{bmatrix} m_1 g & 0 & \dots & m_{10} g & 0 \end{bmatrix} \end{aligned}$$

$$D = \begin{bmatrix} -m_0 \end{bmatrix}$$

The corresponding state-space representation of the system is given by:

$$\begin{cases} \dot{\mathbf{x}}(t) = \mathbf{A}\mathbf{x}(t) + \mathbf{B}u(t) \\ \mathbf{y}(t) = \mathbf{C}\mathbf{x}(t) + \mathbf{D}u(t) \end{cases} \quad (4)$$

2.2. Task 2b: EOM and state-space model with damping

We now want to analyze the model including the damping ratio $\gamma = 0.003$,

Original EOM:

$$m_i \ell_i^2 \ddot{X}_{i2} + m_i g \ell_i X_{i1} = -m_i \ell_i u \quad (5)$$

$$\ddot{X}_{i2} + \frac{g}{\ell_i} X_{i1} = -\frac{1}{\ell_i} u$$

$$\ddot{X}_{i2} + \omega_i^2 X_{i1} = -\frac{1}{\ell_i} u$$

with damping:

$$\ddot{X}_{i2} + 2\gamma\omega_i \dot{X}_{i2} + \gamma^2 X_{i1} = -\frac{1}{\ell_i} u$$

$$\ddot{X}_{i2} = -\omega_i^2 X_{i1} - 2\gamma\omega_i \dot{X}_{i2} - \frac{1}{\ell_i} u \quad (6)$$

only the matrix A will be affected by the changes made as the damping ratio γ does not appear in any other matrix; B comprehends only the EOM derived by the input u , C and D describe the output force where the damping ratio is not included in the formula.

$$A = \begin{bmatrix} 0 & 1 & 0 & 0 & \cdots & 0 & 0 \\ -\omega_1^2 & -2\gamma\omega_1 & 0 & 0 & \cdots & 0 & 0 \\ 0 & 0 & 0 & 1 & \cdots & 0 & 0 \\ 0 & 0 & -\omega_2^2 & -2\gamma\omega_2 & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & 0 & \cdots & 0 & 1 \\ 0 & 0 & 0 & 0 & \cdots & -\omega_{10}^2 & -2\gamma\omega_{10} \end{bmatrix}$$

3. Task 3: Impulse response

This section aims to compute the time response of the net force $F_x(t)$ acting on a tank following an impulsive acceleration, using a reduced-order model with only the first sloshing mode ($n = 1$) and a damping ratio $\zeta = 0.003$.

The response is obtained from 0 to 10 s using four methods: the MATLAB[®] built-in impulse function, a direct analytical solution, the analytical modal form, and a numerical integration technique. The results are compared to assess consistency and highlight differences among the approaches.

3.1. Method 1 – MATLAB[®] impulse function

This method utilizes the MATLAB[®] built-in *ss()* and *impz()* functions to numerically compute the system response to a unit impulse.

The state-space model was defined with the following:

$$\mathbf{A}_n = \begin{bmatrix} 0 & 1 \\ -\omega_n^2 & -2\gamma\omega_n \end{bmatrix}, \quad \mathbf{B}_n = \begin{bmatrix} 0 \\ -\frac{1}{L_n} \end{bmatrix}, \quad \mathbf{C}_n = \begin{bmatrix} m_n g & 0 \end{bmatrix}, \quad \mathbf{D}_n = -m_0$$

The impulse response accurately matched the expected oscillatory behavior and served as a reference for comparison with other methods.

3.2. Method 2 – Analytical solution

The closed-form solution to the damped second-order differential equation with an impulse input was derived using Laplace transforms.

Analytical Derivation of the Impulse Response for the Fundamental Sloshing Mode

We consider the sloshing motion modeled as a single-degree-of-freedom damped oscillator, governed by the following equation:

$$\ddot{\theta}(t) + 2\gamma\omega\dot{\theta}(t) + \omega^2\theta(t) = -\frac{1}{L}\delta(t) \quad (7)$$

where:

- γ is the damping ratio;
- $\omega = \sqrt{g/L}$ is the undamped natural frequency;
- L is the equivalent pendulum length of the first sloshing mode;
- $\delta(t)$ is the Dirac delta function modeling an impulsive tank acceleration;
- Initial conditions: $\theta(0^-) = 0$, $\dot{\theta}(0^-) = 0$.

Laplace Transform

Taking the Laplace transform of both sides (with zero initial conditions):

$$\mathcal{L}\left\{\ddot{\theta} + 2\gamma\omega\dot{\theta} + \omega^2\theta\right\} = \mathcal{L}\left\{-\frac{1}{L}\delta(t)\right\}$$

$$s^2\Theta(s) + 2\gamma\omega s\Theta(s) + \omega^2\Theta(s) = -\frac{1}{L}$$

Solving for $\Theta(s)$:

$$\Theta(s) = -\frac{1}{L} \cdot \frac{1}{s^2 + 2\gamma\omega s + \omega^2} \quad (8)$$

Inverse Laplace Transform

We define the damped natural frequency:

$$\omega_d = \omega\sqrt{1 - \gamma^2} \quad (9)$$

The inverse Laplace transform of the second-order system is known:

$$\mathcal{L}^{-1} \left\{ \frac{1}{s^2 + 2\gamma\omega s + \omega^2} \right\} = \frac{1}{\omega_d} e^{-\gamma\omega t} \sin(\omega_d t)$$

Thus, the impulse response is:

$$\theta(t) = \frac{1}{L\omega_d} e^{-\gamma\omega t} \sin(\omega_d t)$$

Therefore, the analytical expression becomes:

$$\boxed{\theta(t) = \frac{1}{\omega_d L} e^{-\gamma\omega t} \sin(\omega_d t), \quad \text{with} \quad \omega_d = \omega\sqrt{1 - \gamma^2}} \quad (10)$$

Impulse Response of the Net Force $F_x(t)$

We want to compute the time history of the net force $F_x(t)$ exerted by the fluid on the tank structure, under an impulsive acceleration. To obtain the net force $F_x(t)$, we recall that the sloshing mass m exerts a horizontal force on the tank due to its gravitational force and its lateral displacement, which can be approximated as:

$$F_x(t) = mg \sin \theta(t) \approx mg \theta(t) \quad (11)$$

This is the analytical expression for the net force $F_x(t)$ exerted by the sloshing liquid on the tank in terms of the angular response $\theta(t)$.

The force is then calculated as $F_x(t) = mg \theta(t)$, showing near perfect agreement with the numerical impulse response of MATLAB.

3.3. Method 3 – Analytical modal form

In this approach, the problem is formulated in terms of modal coordinates, using the eigenstructure of the system.

For a single mode, this simplifies to the same expression as in Method 2.

However, this formulation sets the stage for generalization to higher-order modal expansions.

3.4. Method 4 – Numerical integration

We used an equivalent initial condition to set the impulse: $x(0) = 0, \quad \dot{x}(0) = \frac{1}{L}$

This effectively embeds the impulse effect into the initial state of the system.

To understand what integration method to select we performed a direct comparison between `ode45` and `ode113`.

`ode45` is based on an explicit 4/5 order Runge-Kutta method, efficient for problems with smooth solutions and moderate stiffness.

`ode113` is a multi-step solver with variable order, typically used for problems requiring high accuracy over long time spans.

Although `ode113` offers potentially higher-order accuracy, the comparison with the analytical solution shows that `ode45` more accurately captures the system impulse response. This is consistent with the nature of `ode45`, which is better suited to handling localized dynamic variations. Therefore, `ode45` is preferred

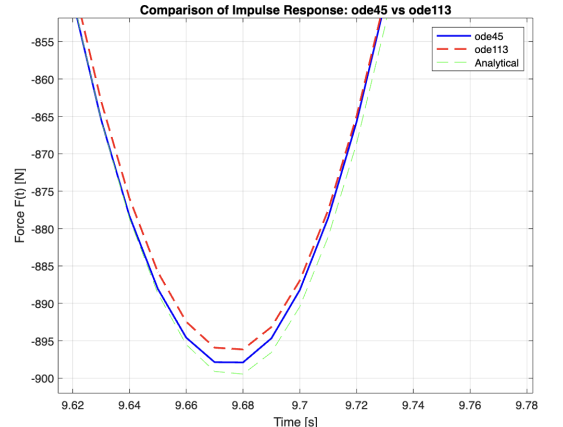


Figure 3: Integration Method Comparison

in this case for both its local precision and ease of implementation.

The resulting system of ODEs was integrated using `ode45`. The output displacement $\theta(t)$ was post-processed to compute $F_x(t) = mg\theta(t)$, yielding results in close agreement with the analytical and impulse-based methods.

3.5. Results and discussion

All methods were implemented using the same time discretization vector, defined as $t = 0:0.01:10$ s to ensure consistency in the comparison of the impulse response.

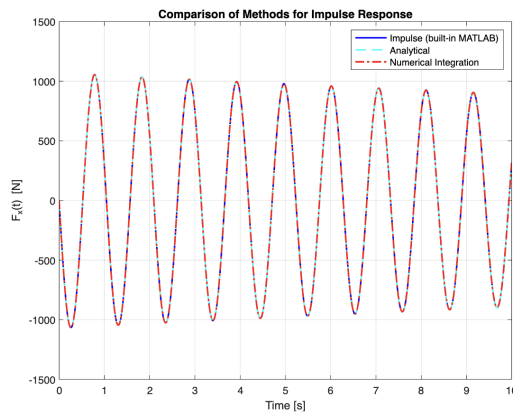


Figure 4: Response Comparison

The built-in MATLAB[®] impulse function yields a time history for the force $F_x(t)$ that is virtually indistinguishable from the analytical solution, confirming its accuracy and reliability in modeling linear time-invariant systems under impulsive excitation.

On the other hand, the numerical solution obtained using `ode45`, although generally in good agreement with the other methods, presents small but noticeable deviations.

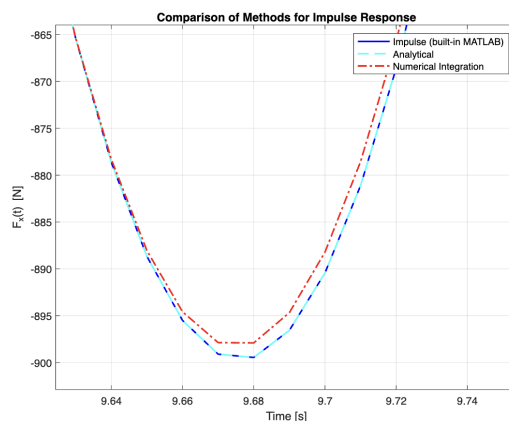


Figure 5: Detailed comparison of the responses (zoomed view)

Conclusion

In summary, while all methods provide a valid estimation of the system response, the analytical and built-in impulse solutions are more precise under idealized conditions, whereas numerical integration introduces minor inaccuracies inherent to its algorithmic implementation.

4. Task 4: Accuracy of the multi-pendulum model

We performed an iteration process to determinate the required model order n for the damped multi-pendulum system to ensure sufficient accuracy in the evaluation of the net force F_x induced by an impulsive acceleration of the tank.

Due to the linearity of the system, the total response is obtained by superposing the individual impulse responses of each pendulum, obtained by the built-in MATLAB[®] impulse function.

Convergence is assessed through the maximum relative error between the peaks amplitudes of the force response $F_x(t)$ for successive iterations as

$$e_{rel} = \max \left| \frac{y_{n,max} - y_{n-1,max}}{y_{n-1,max}} \right| \quad (12)$$

with the termination criterion set at $\epsilon = 0.1\%$.

First, for each n , the total force response was computed by summing the impulse response of each individual pendulum. Therefore, the time vector was partitioned into 20 subintervals, each centered around a peak, based on the half-period $T/2$ of the first pendulum response. These intervals were stored in the vector `i_vec`.

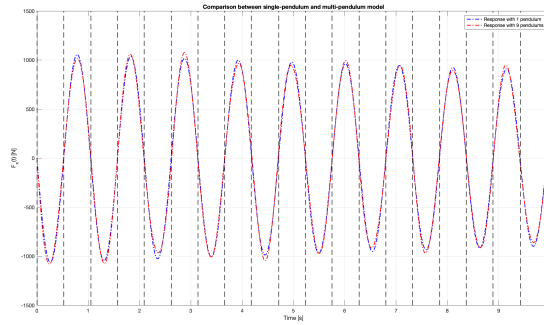


Figure 6: Response Comparison with Interval Partition

In each subinterval, for both the current (n) and the previous ($n - 1$) configuration, we calculated the absolute maximum of the force response.

Then, after calculating the relative error for each peak $e_{rel,i} = \left| \frac{y_{n,max,i} - y_{n-1,max,i}}{y_{n-1,max,i}} \right|$ with $i = 1, \dots, 19$ (number of peaks), we selected the maximum among these errors as the representative error ϵ_n between configurations with n and $n - 1$ pendulums.

This value was stored in the vector `err_rel`.

As soon as the error dropped below a predefined threshold $\epsilon = 0.1\%$, the iteration was terminated.

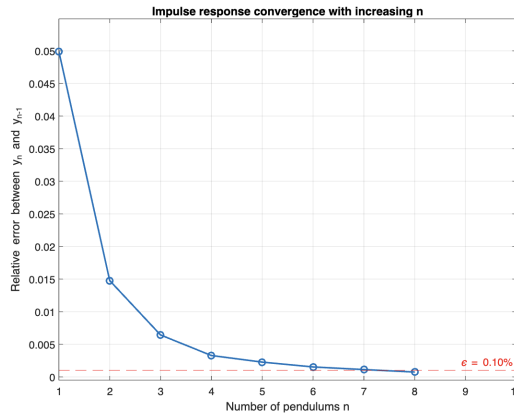


Figure 7: Relative Error

The evolution of the response with increasing n is shown in Fig. 6, while the corresponding relative error ϵ_n is reported in Fig. 7.

To further refine the decision, the derivative of the relative error was also computed. This metric quantifies the percentage reduction in the error from $n - 1$ to n pendulums, providing insight into the value returns of adding more pendulums.

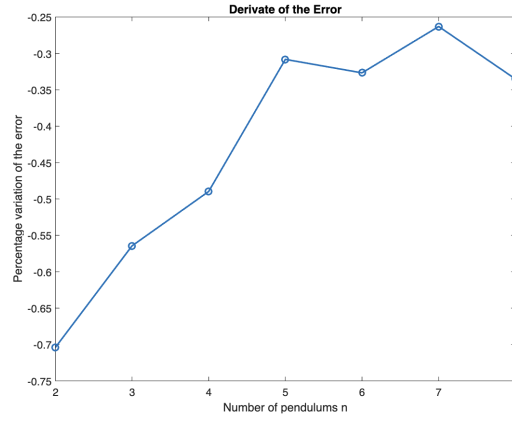


Figure 8: Percentage Error Variation

Initially, adding 1 pendulum, the error dropped by 71%, using 3 pendulums the error decreased by another 56%.

Although the convergence criterion $e_{rel} < \varepsilon$ was met for $n = 7$, the error reduction beyond $n = 5$ was observed to be nearly constant (see Fig. 8).

In fact, the maximum relative difference calculated between $n=7$ and $n=5$ is 0.075%

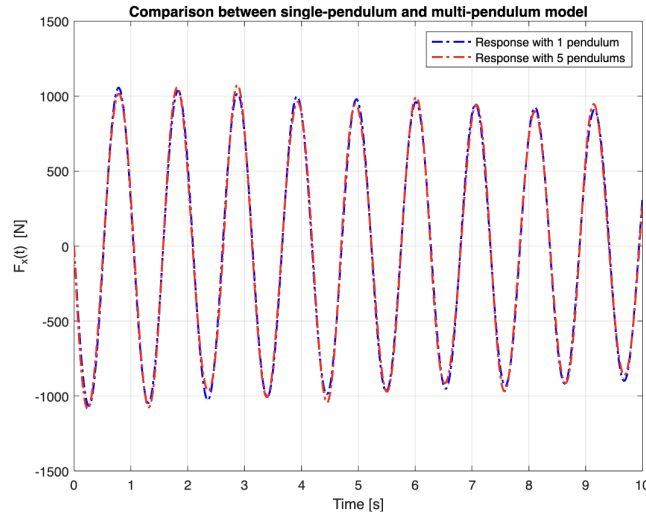


Figure 9: Response Comparison for $n=1$ and $n=5$

As a result, $n = 5$ was selected as the optimal value, offering a good trade-off between accuracy and computational efficiency, providing a relative error between successive iterations of $e_{rel} = 0.23\%$ and, most impressive, a relative error between y_1 (for $n=1$) and y_5 (for $n=5$) of 4.3%

5. Task 5: Step response (MATLAB[®] code)

5.1. Method 1 - MATLAB[®] impulse function

In this approach, the net horizontal force $F_x(t)$ due to a step acceleration of the tank is computed using the built-in `step` function in MATLAB[®]. The system under consideration is a damped multi-pendulum model of order $n = 5$, where each pendulum represents a mode of sloshing behavior within the fluid.

Given the linearity of the model, the superposition principle is employed.

Thus, the total force $F_x(t)$ is calculated as the sum of the individual contributions from each pendulum. The fixed mass m_0 is modeled separately and added once as a constant term, while the dynamic contributions from the sloshing modes are computed iteratively.

The time vector spans from 0 to 100 seconds with 1000 equally spaced points. For each pendulum n , a second-order state-space model is defined as follows:

$$\mathbf{A}_n = \begin{bmatrix} 0 & 1 \\ -\omega_n^2 & -2\gamma\omega_n \end{bmatrix}, \quad \mathbf{B}_n = \begin{bmatrix} 0 \\ -\frac{1}{L_n} \end{bmatrix}, \quad \mathbf{C}_n = \begin{bmatrix} m_n g & 0 \end{bmatrix}, \quad \mathbf{D}_n = 0$$

where ω_n is the natural frequency of the n -th pendulum, γ is the damping ratio (assumed equal for all modes), L_n is the pendulum length, and m_n is the modal mass. The output of interest is the horizontal force contribution of each pendulum due to the step input acceleration.

The overall net force is initialized as:

$$F_{x_0}(t) = -m_0 u(t) \quad (\text{constant contribution from the rigid mass with unit step acceleration})$$

where $u(t)$ is the unit step function.

Then, for each mode n , the state-space system is created and the response to a unit step input is obtained via:

$$[\mathbf{y}, \mathbf{tOut}] = \text{step}(\text{sys}, \mathbf{t});$$

Finally, the contributions are accumulated to form the complete force history:

$$F_x(t) = -m_0 + \sum_{n=1}^7 F_{x,n}(t) \quad (13)$$

This method leverages the efficiency and robustness of MATLAB[®] internal solver for linear time-invariant systems, and provides a fast and accurate estimation of the system dynamic response to the input acceleration.

5.2. Method 2 - Analytical solution

This section presents the analytical solution of the system response due to a unit step acceleration input, focusing on the evolution of the angular displacement $\theta(t)$ for each sloshing pendulum. The pendulum dynamics are governed by the second-order differential equation:

$$\ddot{\theta} + 2\gamma\omega_0\dot{\theta} + \omega_0^2\theta = -\frac{1}{L} \cdot u \quad (14)$$

Assuming a unit step input acceleration, i.e. $u = 1$, and zero initial conditions (rest initial state), the equation becomes:

$$\ddot{\theta} + 2\gamma\omega_0\dot{\theta} + \omega_0^2\theta = -\frac{1}{L}$$

This is a second-order linear differential equation with constant coefficients and a constant forcing term. Its general solution consists of the homogeneous solution plus the particular solution:

$$\begin{aligned} \theta(t) &= \theta_{\text{hom}}(t) + \theta_{\text{part}} \\ \theta_{\text{part}} &= -\frac{1}{\omega_0^2 L} \end{aligned}$$

For $\gamma < 1$, the homogeneous solution is:

$$\theta_{\text{hom}}(t) = e^{\sigma t} (A \cos(\omega t) + B \sin(\omega t))$$

where:

$$\begin{aligned}\sigma &= -\gamma\omega_0 \\ \omega &= \omega_0\sqrt{1 - \gamma^2}\end{aligned}$$

To determine constants A and B , we apply the initial conditions:

$$\theta(0) = 0 \quad \text{and} \quad \dot{\theta}(0) = 0$$

At $t = 0$:

$$\theta(0) = A - \frac{1}{\omega_0^2 L} = 0 \quad \Rightarrow \quad A = \frac{1}{\omega_0^2 L}$$

Then, differentiating $\theta(t)$ and evaluating at $t = 0$:

$$\begin{aligned}\dot{\theta}(0) &= \sigma A + B\omega = 0 \\ \Rightarrow B &= -\frac{\sigma A}{\omega} = \frac{\gamma}{\omega\omega_0 L}\end{aligned}$$

Replacing all terms, the full time-domain expression for $\theta(t)$ becomes:

$$\boxed{\theta(t) = \frac{1}{L} e^{\sigma t} \left(\frac{\cos(\omega t)}{\omega_0^2} + \frac{\gamma \sin(\omega t)}{\omega\omega_0} \right) - \frac{1}{\omega_0^2 L}} \quad (15)$$

The horizontal force contribution from the n -th mode is then:

$$F_{x,n}(t) = m_n g \cdot \theta_n(t)$$

The total net force is obtained by summing the modal contributions and the effect of the static mass m_0 :

$$F_x(t) = -m_0 + \sum_{n=1}^7 F_{x,n}(t)$$

This solution was implemented numerically in MATLAB[®] for each sloshing mode. The precision of this analytical expression provides a reliable benchmark for validating both simulation-based and numerical integration results.

5.3. Method 3 - Numerical integration

In this approach, the net force $F_x(t)$ is computed by numerically integrating the equations of motion for each sloshing pendulum using a time-domain integration scheme. The second-order differential equation governing each mode is given by:

$$\ddot{\theta} + 2\gamma\omega_0\dot{\theta} + \omega_0^2\theta = -\frac{1}{L} \cdot u$$

Assuming a unit step acceleration input, i.e., $u(t) = 1$, the equation becomes:

$$\ddot{\theta}(t) = -2\gamma\omega_0\dot{\theta}(t) - \omega_0^2\theta(t) - \frac{1}{L}$$

This second-order ordinary differential equation (ODE) is reformulated as a first-order system for numerical integration:

$$\begin{cases} \dot{x}_1 = x_2 \\ \dot{x}_2 = -2\gamma\omega_0x_2 - \omega_0^2x_1 - \frac{1}{L} \end{cases} \quad \text{with} \quad x(0) = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

The MATLAB[®] function `ode45` is employed to solve this system. This function is based on an explicit Runge-Kutta (4,5) formula, the Dormand-Prince pair, which is well-suited for problems that are non-stiff and require adaptive step sizing to ensure both accuracy and computational efficiency.

The use of `ode45` is justified as follows:

- The system is smooth and does not exhibit stiffness.
- Adaptive step-size control offers improved performance compared to fixed-step methods.
- It is a robust general-purpose solver widely adopted for dynamic systems with moderate complexity.

The code integrates each pendulum dynamics individually. The angular displacement $\theta_n(t)$ is then used to compute the modal contribution to the net horizontal force:

$$F_{x,n}(t) = m_n g \cdot \theta_n(t)$$

Finally, the overall force response is obtained by summing the modal contributions and the static component of the mass m_0 :

$$F_x(t) = -m_0 + \sum_{n=1}^7 F_{x,n}(t)$$

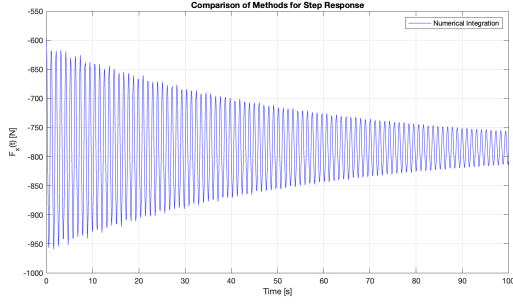
This numerical approach offers a flexible and accurate method for simulating the system response, especially when closed-form solutions are intractable or when external inputs are non-analytical.

5.4. Results and discussion

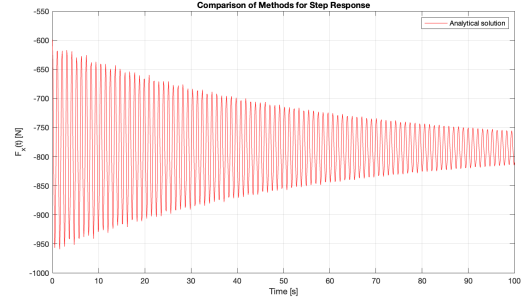
To validate the consistency and accuracy of the modeling approach, the net force $F_x(t)$ resulting from a unit step acceleration was computed using three different methods:

1. the built-in `step` function provided by MATLAB[®],
2. the analytical solution of the damped oscillator,
3. and a numerical integration scheme based on `ode45`.

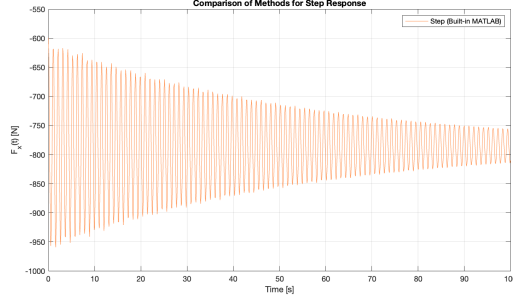
The resulting time histories are plotted below for visual comparison:



(a) Net force response $F_x(t)$ obtained via numerical integration using the `ode45` solver.



(b) Net force response $F_x(t)$ derived from the analytical expression of the system.



(c) Net force response $F_x(t)$ computed using the built-in MATLAB `step` function.

Figure 10: The three methods compared

We can generate the responses with 100 equally spaced points instead of 1000, and overlap the graphics:

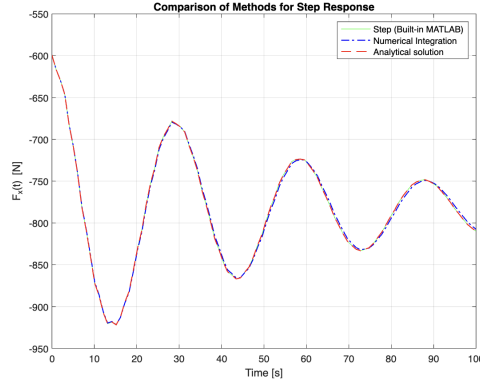


Figure 11: Direct comparison between the three methods

The three response curves are in excellent agreement with each other, demonstrating:

- Identical oscillation frequencies, as governed by the damped natural frequency ω_d ;
- Matching decay rates due to the shared damping ratio γ ;
- A common steady-state force level as $t \rightarrow \infty$, confirming that all transient components diminish consistently across methods.

While slight differences in numerical precision or resolution may be present, they are negligible in practical terms and do not compromise the physical validity of the results.

Conclusion

The observed consistency confirms the correctness of the system modeling and implementation. Each method successfully captures the transient and steady-state behavior of the net force induced by a step acceleration. As a result, any of the three methods may be adopted depending on the application context whether analytical tractability, simulation speed, or numerical flexibility is prioritized.

6. Task 6 : Frequency response

It is requested to derive a frequency response function corresponding to a damped n -order multi-pendulum model. We will select $n = 5$ according to the analysis done in **Task 4**.

The key difference from previous tasks is the input value: rather than applying a lateral acceleration to the tank, the input variable $u(t)$ is now defined as the position of the tank.

The state-space representation of the system is now given by:

$$\begin{cases} \dot{\mathbf{x}}(t) = \mathbf{A}\mathbf{x}(t) + \mathbf{B}\ddot{u}(t) \\ \mathbf{y}(t) = \mathbf{C}\mathbf{x}(t) + \mathbf{D}\ddot{u}(t) \end{cases} \quad (16)$$

Applying the Laplace transform and assuming the initial conditions to be equal to zero:

$$s\mathbf{X}(s) = \mathbf{A}\mathbf{X}(s) + \mathbf{B}s^2U(s)$$

This leads to:

$$\mathbf{X}(s) = (s\mathbf{I} - \mathbf{A})^{-1}\mathbf{B}s^2U(s)$$

and the output becomes:

$$\mathbf{Y}(s) = \mathbf{C}\mathbf{X}(s) + \mathbf{D}s^2U(s)$$

$$\Rightarrow \mathbf{Y}(s) = [\mathbf{C}(s\mathbf{I} - \mathbf{A})^{-1}\mathbf{B} + \mathbf{D}] s^2U(s)$$

The transfer function $H(s)$ of a linear time-invariant (LTI) system is defined as the fraction of the Laplace transform of the output and the Laplace transform of the input:

$$H(s) = \frac{Y(s)}{U(s)}$$

Therefore, considering the system is moved by the acceleration input $\ddot{u}(t)$, the overall transfer function is:

$$H(s) = [\mathbf{C}(s\mathbf{I} - \mathbf{A})^{-1}\mathbf{B} + \mathbf{D}] s^2 \quad (17)$$

For comparison, we will also consider a frozen liquid model, which assumes that the entire mass of the fluid moves rigidly with the tank, without internal deformation or sloshing effects.

The total force exerted by the fluid on the tank corresponds to a reaction force due to its inertia. Since the fluid mass undergoes the same acceleration as the tank, the output is directly proportional to the second derivative of the input:

$$F(t) = m_{\text{tot}} \cdot \ddot{u}(t) \quad (18)$$

Taking the Laplace transform under the initial conditions:

$$F(s) = m_{\text{tot}} \cdot s^2U(s)$$

The transfer function from input position to output force is given by:

$$H_{\text{frozen}}(s) = \frac{F(s)}{U(s)} = m_{\text{tot}} \cdot s^2 \quad (19)$$

where m_{tot} is the total effective fluid mass, including both the base mass (rigidly attached) and the additional sloshing masses (represented by the pendulums) considered frozen in this case.

Now we can write on MATLAB[®] the code to compute the frequency response of the system. The resulting transfer function is evaluated over a range of excitation frequencies and is compared to the response of an equivalent system with a rigid, non-sloshing (frozen) liquid mass. This comparison highlights the dynamic differences between the two models and illustrates the effect of internal liquid motion on the overall system behavior.

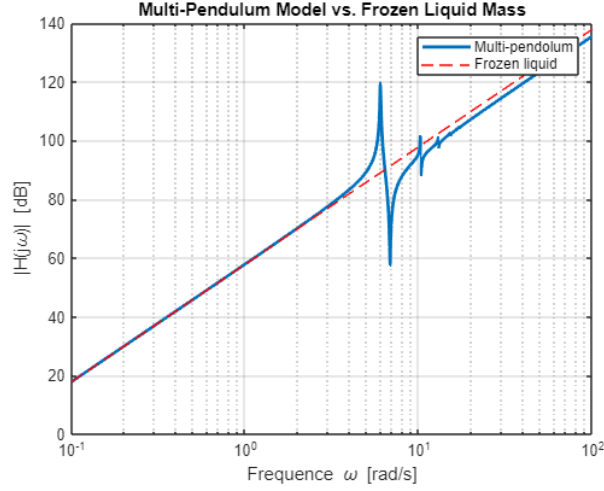


Figure 12: Frequency Response Comparison: Multi-Pendulum Model vs. Frozen Liquid Mass

6.1. Discussion on frequency responses differences

The main differences between the multi-pendulum model and the frozen-liquid model are most evident at mid and high frequencies. At mid frequencies, the pendulum system shows resonant peaks corresponding to the natural frequencies of the individual modes; these arise from the internal dynamics of the liquid, which are not captured in the rigid frozen mass model.

At high frequencies, the response of the pendulums becomes increasingly rigid as they can no longer follow the rapid oscillations. When the base mass is correctly included and the total mass matches that of the frozen model, the two responses tend to converge asymptotically. The remaining differences are due to the limited number of pendulums used to approximate the continuous sloshing behavior.

Both the multi-pendulum and the frozen liquid models exhibit an increasing magnitude at high frequencies. This behavior is due to the fact that the system is accelerated by the second derivative of the input position which is the acceleration of the tank. In the Laplace domain, this corresponds to a multiplication by s^2 that in the frequency domain becomes $(j\omega)^2$. As a result, the magnitude of the transfer function grows proportionally to ω^2 , leading to a slope of $+40\text{dB/dec}$ in the Bode magnitude graph for both models.

7. Task 7: Simulink[®] model

After selecting $n = 5$ as the order for the damped multi-pendulum system of the tank, we implemented a Simulink model to compute the time history of the net force F_x resulting from a step acceleration applied to the tank from 0 to 100 seconds. The simulation parameters were initially set with a start time of 0.0 s and a final time of 100.0 s. A variable-step solver (ode45) was chosen to perform the numerical integration, the same as used in the previous tasks.

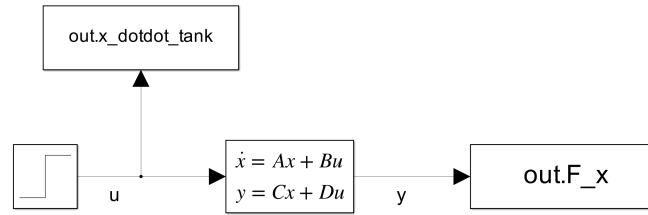
To model the damped n -pendulum system, the following Simulink blocks were implemented:

- a **Step** block
- a **State-Space** block
- and two **To Workspace** blocks (used to export the tank acceleration $\ddot{x}_{\text{tank}}$ and the net force F_x to MATLAB[®])

The Step block was configured with the following parameters: step time = 0 s, initial value = 0, final value = 1, and sample time = 0. The output of this block, representing the acceleration of the tank $\ddot{x}_{\text{tank}}$, was sent to MATLAB[®] via a **To Workspace** block to allow post-processing and plotting alongside the resulting net force F_x .

To compute F_x , a State-Space block was used. The state-space matrices A , B , C , and D were constructed specifically for this model. A **for** loop (from 1 to n_{max}) was used to generate the contribution of each individual pendulum, ensuring that they did not interact with each other. To build the matrix A , we used the MATLAB[®] function `blkdiag`, which returns a block diagonal matrix formed by placing the input matrices $A_1, \dots, A_N$ along the main diagonal. The other matrices (B , C , and D) were assembled by iteratively adding their respective components.

Finally, the computed net force F_x , obtained from the output of the State-Space block, was exported to MATLAB[®] using the second **To Workspace** block. Both $\ddot{x}_{\text{tank}}$ and F_x were saved in array format as 2-D signals for further analysis, plotting and compatibility. Below is the Simulink model implemented, along with the corresponding blocks previously mentioned.



By plotting the response, it is possible to observe that it is consistent with the one obtained from the previous graphs (see Fig. 10) and that the decision to use n pendulums from the analysis of **Task 4** has proven to be correct and accurate for the model.

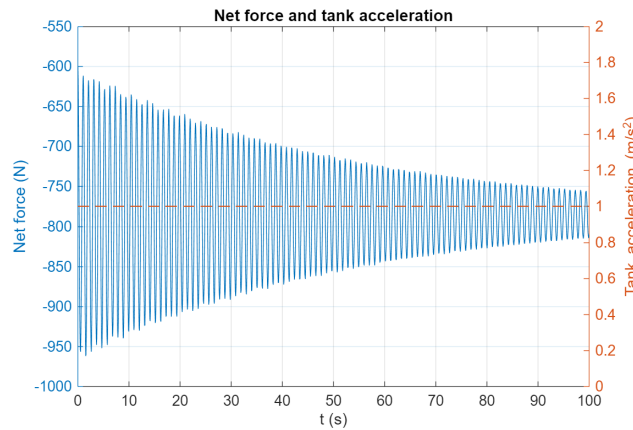


Figure 13: Net Force F_x plot due to a step acceleration of the tank $\ddot{x}_{\text{tank}}$

Conclusion

We successfully implemented and analyzed the Slosh-Mass Pendulum system through analytical modeling and simulation using MATLAB[®] and Simulink[®]. We verified the responses of the model to various inputs, while also evaluating the accuracy and efficiency of different model configurations, regarding the number of pendulums used. This aspect is critical for ensuring convergence and computational efficiency in systems designed to approximate chaotic dynamics, such as those observed in liquid sloshing within tanks.

This approach exemplifies an engineering methodology, which emphasizes the modeling of complex physical systems for real-world applications, while preserving a high level of accuracy and analytical tractability.

References

- [1] F. T. Dodge. *Dynamic Behavior of Liquids in Moving Containers*. Springer, 2005 (cit. on pp. 3, 4).